

Lecture 22

Synchronous Counters

Synchronous binary counters · up-down counters · parallel load
· BCD and arbitrary sequences · ring and Johnson counters

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 6, §6.4–6.5

0	1	1	0
0	1	1	1
1	0	0	0

every bit on the same edge

Today's roadmap

1

§6.4

Binary counters

One clock for every bit

2

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Up-down, load

Count either way; preset

3

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BCD, arbitrary

Any sequence; unused states

4

§6.5

Ring, Johnson

Shift-register counters

By the end of class you can...

- Derive the T inputs of a synchronous binary counter
- Add up-down control and parallel load
- Design a BCD counter and check its unused states
- Design a counter for an arbitrary sequence
- Trace ring and Johnson counters and decode their states

Agenda for today's session



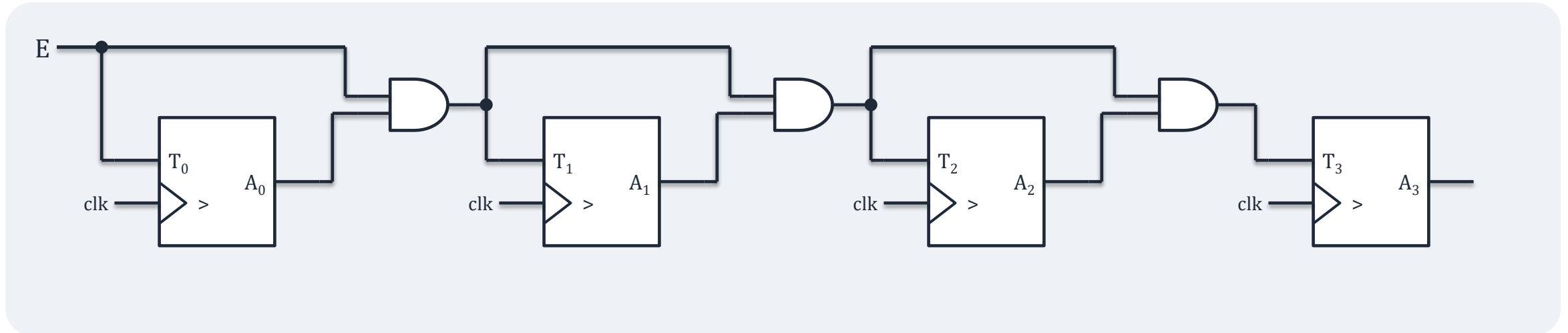
Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homework 10 collected	—
0:05 – 0:25	Synchronous binary counters	Example 1
0:25 – 0:40	Up-down counters	Example 2
0:40 – 0:50	Break	
0:50 – 1:15	Parallel load; the BCD counter	Example 3
1:15 – 1:40	Arbitrary sequences; ring and Johnson	Example 4 · Your turn
1:40 – 1:50	Concept check and wrap-up	Exam 3 preview

Example 1 — A synchronous counter

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EXAMPLE

Design a 4-bit synchronous binary up-counter with T flip-flops and a count enable E .



1 Binary counting: a bit toggles when every lower bit is 1

2 T flip-flops: $T = 1$ means toggle on the next edge

3 Bit 0 toggles on every enabled clock: $T_0 = E$

4 $T_1 = EA_0$, $T_2 = EA_0A_1$, $T_3 = EA_0A_1A_2$

5 Each AND adds one bit to the previous enable

6 One shared clock: no false counts

Example 2 — An up-down counter

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EXAMPLE

3-bit counter with inputs Up and Down (Up wins if both are 1). Find the T inputs.

Up	Down	Action
1	X	count up
0	1	count down
0	0	hold

Down-count rule

Counting down, a bit toggles when every lower bit is 0: the borrow.

- 1 Enables: $U = Up$; $D_n = Up' \text{ Down}$ (Up wins)
- 2 Bit 0 toggles on every up or down count: $T_0 = U + D_n$
- 3 Up: bit 1 toggles when $A_0 = 1 \rightarrow UA_0$
- 4 Down: bit 1 toggles when $A_0 = 0$ (a borrow) $\rightarrow D_n A'_0$
- 5
$$T_1 = UA_0 + D_n A'_0$$
- 6 Same rule for bit 2: $T_2 = UA_0 A_1 + D_n A'_0 A'_1$
- 7 Checked on all 32 cases of Up, Down and state ✓

Break

10 minutes — we resume at the 0:50 mark

Next: counters with load, BCD and other sequences

Counters with parallel load

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Control inputs

Clear: force 0000. Load: copy the inputs $I_3 \dots I_0$. Count: add 1. Otherwise hold.

Each stage chooses its next value with a multiplexer (as in Lecture 21), not by gating the clock.

Build any modulus

Load 0000 when the count reaches $N - 1$: a mod- N counter.

Mod 10: detect 1001 with $Q_3 Q_0$ and assert Load with inputs 0000.

Clear	Load	Count	Next
1	X	X	0000
0	1	X	$I_3 I_2 I_1 I_0$
0	0	1	$A + 1$
0	0	0	A (hold)

A synchronous clear or load acts at the next clock edge.

An asynchronous clear acts immediately and can cause a glitch of the unwanted state.

Example 3 — The BCD counter

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EXAMPLE

Design a synchronous BCD counter (0–9) with T flip-flops and output $y = 1$ at count 9. Check the unused states.

$Q_8Q_4Q_2Q_1$	next	$T_8T_4T_2T_1$
0000	0001	0001
0001	0010	0011
0010	0011	0001
0011	0100	0111
0100	0101	0001
0101	0110	0011
0110	0111	0001
0111	1000	1111
1000	1001	0001
1001	0000	1001

1 Maps with 10–15 as don't-cares give:

2 $T_{Q1} = 1, T_{Q2} = Q_8'Q_1, T_{Q4} = Q_2Q_1$

3 $T_{Q8} = Q_8Q_1 + Q_4Q_2Q_1, y = Q_8Q_1$

4 Unused states, from these equations:

from	10	11	12	13	14	15
to	11	6	13	4	15	2

Every unused state reaches the valid count within two clocks: self-correcting.

Example 3 — Step by step

§6.5

- 1 For each count, $T = \text{present} \oplus \text{next}$, bit by bit
- 2 Map each T column; 10–15 are don't-cares
- 3 T_{Q_1} is 1 on every row $\rightarrow T_{Q_1} = 1$
- 4 $T_{Q_2} = 1$ on 0001, 0011, 0101, 0111 $\rightarrow Q'_8 Q_1$
- 5 $T_{Q_4} = 1$ on 0011, 0111 $\rightarrow Q_2 Q_1$
- 6 $T_{Q_8} = 1$ on 0111, 1001 $\rightarrow Q_8 Q_1 + Q_4 Q_2 Q_1$
- 7 $y = 1$ only at 1001; with $X \rightarrow y = Q_8 Q_1$
- 8 Unused states: put 10–15 into the equations
- 9 e.g. 10 (1010): only $T_{Q_1} = 1 \rightarrow 1011 = 11$
- 10 Each unused state reaches a valid count in ≤ 2 clocks

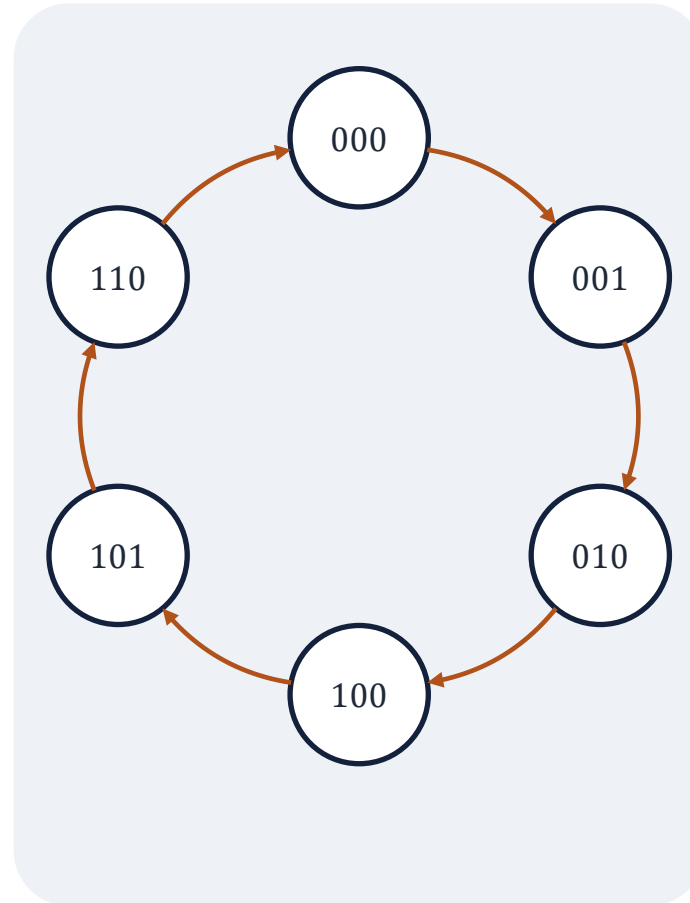
Example 4 — An arbitrary sequence

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EXAMPLE

Design a counter with JK flip-flops that repeats 0, 1, 2, 4, 5, 6. States 3 and 7 are unused.

ABC	next	JA KA	JB KB	JC KC
000	001	0 X	0 X	1 X
001	010	0 X	1 X	X 1
010	100	1 X	X 1	0 X
100	101	X 0	0 X	1 X
101	110	X 0	1 X	X 1
110	000	X 1	X 1	0 X



1 $J_A = K_A = B$

2 $J_B = C, K_B = 1$

3 $J_C = B', K_C = 1$

4 Unused: 011 $\rightarrow$ 100, 111 $\rightarrow$ 000

5 Both recover ✓

Example 4 — Step by step

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- 1 Next state for each present state; 3 and 7 are don't-cares
- 2 For each bit, J, K from the excitation table (bit $\rightarrow$ next bit)
- 3 $J_A = 1$ only at 010 ($\rightarrow$ 100); X where $A = 1 \rightarrow J_A = B$
- 4 $K_A = 1$ at 110 ($\rightarrow$ 000) $\rightarrow K_A = B$
- 5 $J_B = 1$ at 001, 101 $\rightarrow J_B = C$; every K_B is 1 or X $\rightarrow K_B = 1$
- 6 $J_C = 1$ at 000, 100 $\rightarrow J_C = B'$; $K_C = 1$
- 7 Unused 011: $A: JK = 11 \rightarrow 1$; $B: 11 \rightarrow 0$; $C: 01 \rightarrow 0$
- 8 So 011 $\rightarrow$ **100**
- 9 Unused 111: A toggles $\rightarrow 0$; B toggles $\rightarrow 0$; C resets $\rightarrow 0$
- 10 So 111 $\rightarrow$ **000**: both recover ✓

Ring and Johnson counters

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Ring counter: $Q_0 \rightarrow D_3$

Clock	$Q_3Q_2Q_1Q_0$
0	1000
1	0100
2	0010
3	0001

4 flip-flops, 4 states, one hot bit: no decoder needed.

Johnson counter: $Q'_0 \rightarrow D_3$

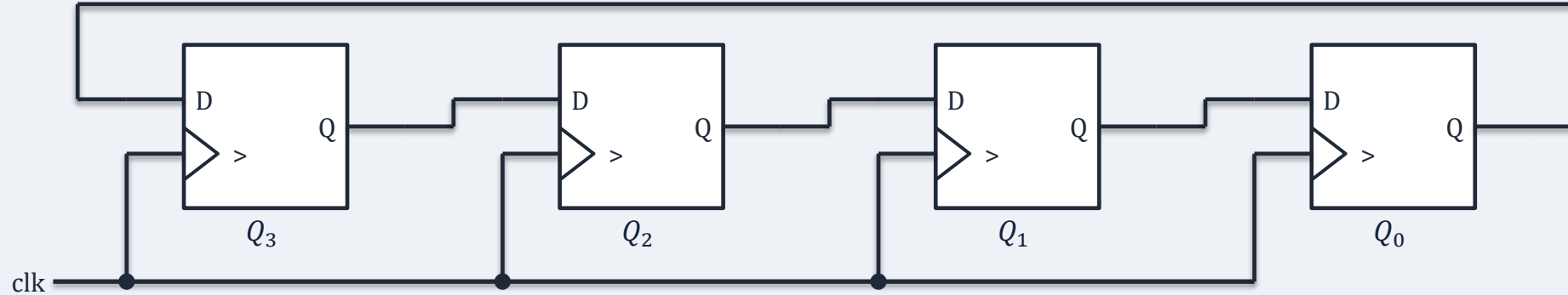
Clock	$Q_3Q_2Q_1Q_0$	Decode
0	0000	$A'E'$
1	1000	AB'
2	1100	BC'
3	1110	CE'
4	1111	AE
5	0111	$A'B$
6	0011	$B'C$
7	0001	$C'E$

4 flip-flops, 8 states; each state decoded by one 2-input AND.

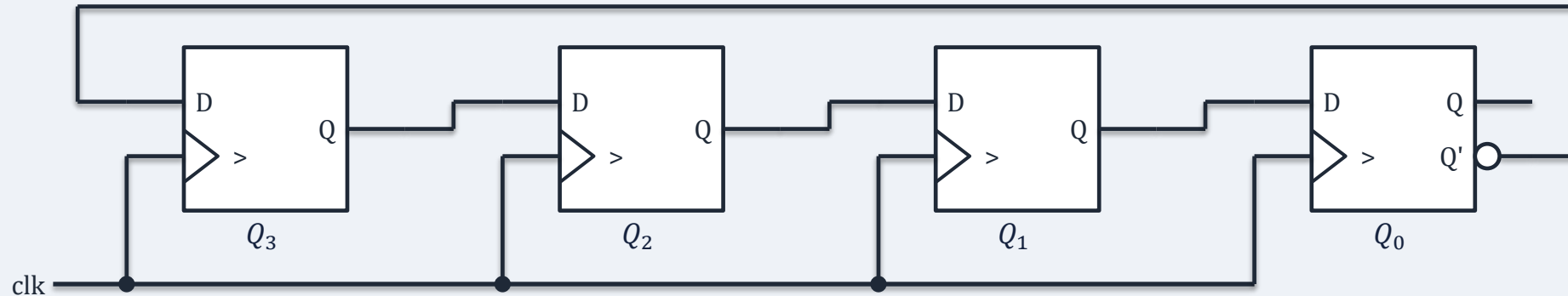
Ring and Johnson circuits

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Ring: Q_0 feeds back



Johnson: Q'_0 feeds back



Your turn — Up-down with D flip-flops

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YOUR TURN

≈ 12 min

A 2-bit counter A_1A_0 counts up when $U = 1$ and down when $U = 0$.

- (a) Write the state table ($U, A_1, A_0 \rightarrow$ next state).
- (b) Find D_1 and D_0 with K-maps. Can you write D_1 with XOR gates?
- (c) How many states does a 5-flip-flop Johnson counter have?

Map variables in the order U, A_1, A_0 .

Your turn — Solution

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SOLUTION

U	A ₁ A ₀	next
0	00	11
0	01	00
0	10	01
0	11	10
1	00	01
1	01	10
1	10	11
1	11	00

		A ₁ A ₀			
		00	01	11	10
U	0	1		1	
	1		1		1

D_1

- 1 (a) $U = 1$: next = present + 1; $U = 0$: present - 1
- 2 (b) D flip-flops: D = next-state bit
- 3 A_0 flips on every clock: $D_0 = A_0'$
- 4 D_1 map: 1s at m0, m3, m5, m6
- 5 A checkerboard: no two 1s adjacent $\rightarrow$ XOR form
- 6
$$D_1 = A_1 \oplus A_0 \oplus U'$$
- 7 Two XOR gates and an inverter
- 8 (c) Johnson: $2n = \mathbf{10}$ states

Concept check

1

Why does a synchronous counter show no false counts?

2

When does bit i toggle in an up-counter? In a down-counter?

3

Why check the unused states of a BCD counter?

4

How many states do ring and Johnson counters with n flip-flops have?

Key takeaways

Binary

$$T_i = E A_0 \cdots A_{i-1}$$

Down

toggle when lower bits are 0

Mod N

load 0 at count $N - 1$

BCD

$$T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1$$

Arbitrary

excitation table + X's

Ring / Johnson

$n / 2n$ states

Same clock, every bit: synchronous counters count cleanly.

Homework #11 due at the start of Lecture 28

This afternoon's lab: Lab Hp4 — Calculator 4: BCD to seven-segment display

Next: review for Exam 3